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Transformation Cleaning

Data Preparation & Quality Notes

Transformation cleaning cleans the dataset to fix quality and consistency issues.

A. Addressing Null / Missing Values

Missing values occur when a field in a record has no recorded data. Left untreated, they can bias averages, break calculations, and reduce the reliability of any downstream analysis. There are two broad ways to address them:

1. Filter out records with missing/null fields.

   This removes the incomplete record entirely. It is simple and safe, but reduces the size of the dataset — useful only when missing records are few and non-critical.

2. Imputation — replacing the missing values using:

   • **Statistical average:** inserting the mean or median of the field.

   e.g. a missing 'age' value is replaced with the average age of all customers.

   • **Similar records:** generating values for 'like' customers.

   e.g. a missing income value is estimated using customers with a similar profile (location, occupation, spending pattern).

   • **Time series logic:** using the last valid value in a sequence.

   e.g. a missing day's stock price is carried forward from the previous day's closing value.

B. Addressing Invalid Values

Invalid values are caused by inconsistency and ambiguity in how data was entered or captured — for example, a negative age, a date in the future, or a percentage above 100. Two remedies are commonly used:

1. Overwrite with calculated / consistent values.

   The invalid entry is replaced with a value derived from business logic or related fields, so the record remains usable.

2. Put it on record but mark it invalid, so that two parallel analyses can take place.

   This keeps the original data intact while flagging it. Running the analysis with and without the flagged records shows how much these invalid values are skewing the final result — giving a clearer picture of their real impact on the data.

C. Data Standardisation

Standardisation ensures that data referring to the same real-world value is represented in one consistent format, so records are not incorrectly treated as different when they should match.

• **Edit distance:** treating similar strings as the same value by measuring how many character edits (insertions, deletions, substitutions) separate them.

e.g. "California" and "Californa" are recognised as the same state despite the spelling error.

• **Domain knowledge:** using external, real-world logic to correctly resolve ambiguous entries.

e.g. using area codes, ZIP codes, or IP addresses to correctly classify an ambiguous entry like "India (North/South)" into the right region.

Key Takeaway — Transformation cleaning makes a dataset trustworthy: fill or remove missing values, correct or flag invalid ones, and standardise inconsistent entries so every downstream calculation reflects reality, not data-entry noise.